Extra homework 2

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This report constructs the product space from country-product export data for 2018 and 2021, following Hidalgo et al. (2007): relatedness is the RCA-based co-occurrence of products across countries, and the networks are filtered to the maximum spanning tree plus all links with proximity ≥ 0.55. I visualize and describe the 2021 network and then compare it with 2018 to assess whether strong links persist.



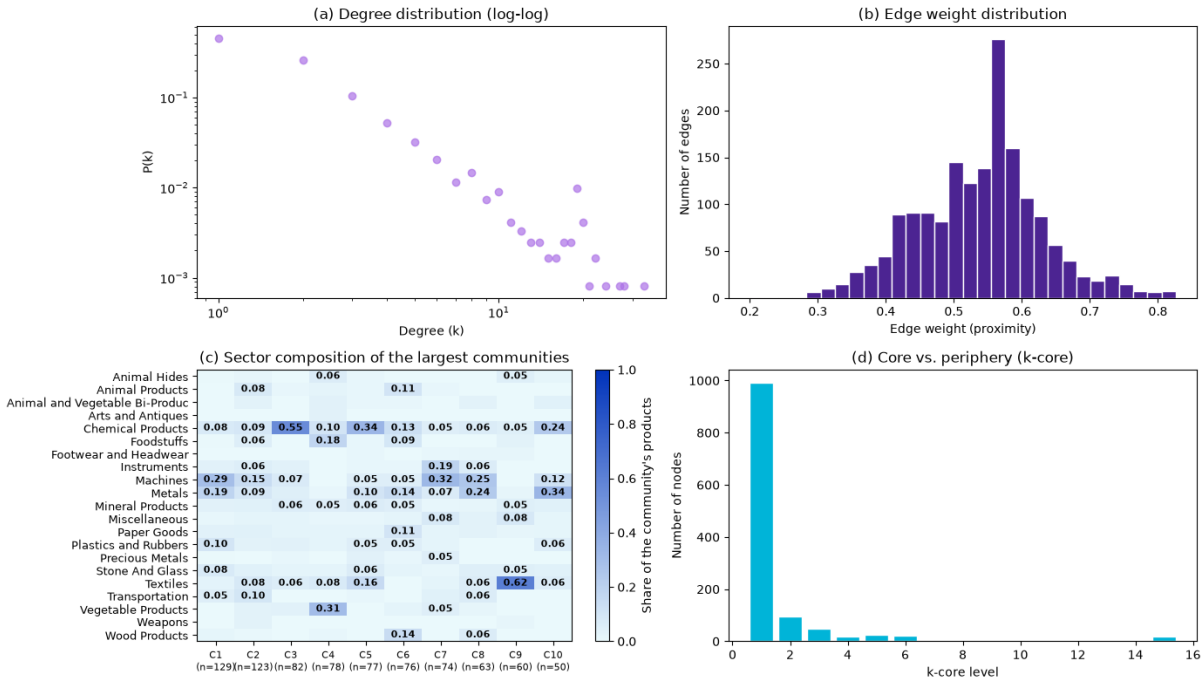
1. Figure: Product Space Relatedness Network Based on 2021 Export Values

The first figure shows the network of export products in 2021. The Textiles (magenta) section forms a clearly recognizable, long chain extending downward from the center of the network and also appears on the left-hand branches; the Machines (dark green) section is concentrated in the lower right corner, where the largest nodes are also located, while the Chemical Products (light orange) section is concentrated in the lower left part and connects to the rest of the network via a thin, long edge.

The central part is highly mixed (Metals, Instruments, Machines, Mineral Products, and Foodstuffs are all next to each other), meaning that here the network structure cuts across the official section boundaries.

In the scatter plot shown in Figure 2/a, we can see that the log-log distribution of degree numbers is heavily skewed to the right. Many nodes have low degree numbers, and there are few hubs in the network.

Histogram 2/b shows the weight distribution of the edges. The edge weights range from approximately 0.28 to 0.82, with the most common values around 0.55-0.6 (approximately 270 edges). This distribution is partly a consequence of the maximum spanning tree method: edges with weights below 0.55 were included due to the spanning tree, while those above 0.55 were included due to the threshold. We can observe that very strong connections above 0.7 are rare.



2. Figure: Structure of the 2021 product space network. (a) Degree distribution (log-log), (b) distribution of edge weights, (c) sector composition of the ten largest communities, (d) k-core distribution.

Based on the 2/c heatmap, the largest clusters consist of 129, 123, 82, and 78 products (C1-C4), while the rest are smaller. Some communities are relatively sector-specific: C9 is predominantly Textiles (0.62), C3 is Chemical Products (0.55), and C4 consists of Vegetable Products and Foodstuff (0.31 and 0.18, respectively), but the majority are mixed (e.g., C1 Machines (0.29) and Metals (0.19), C7 Machines (0.32) and Instruments (0,19)). It follows that the relatedness network partly follows the official HS section classification (textiles, chemicals, agriculture) but intersects the product groups of machinery and metals.

Bar 2/d shows a comparison between the core and the periphery. Most products (nearly 1000) are at k-core level 1, meaning they have a tree-like structure with a sparse periphery. Approximately 100 additional products are at level 2, about 50 at level 3, and only a few at levels 4-6. There are no products at levels 7-14, but there are a few products at level 15. The core is thus sharply distinct: a small, very densely connected group, with no transition from the periphery.

I compared the 2021 network (following the COVID-19 pandemic) with the 2018 network (prior to the COVID-19 pandemic). As can be seen from the first table, the same products appear in both years (the number of vertices is identical), so the two networks are directly comparable, and any differences are not caused by changes in the product set.

The number of edges and the average degree increased slightly, while the number of communities, modularity, and the maximum k-core level decreased slightly. Overall, based on the descriptive statistics, no significant difference can be observed between the two years.

	2018	2021
Nodes	1222	1222
Edges	1685	1728
Average degree	2.76	2.83
Communities (Louvain)	44	38
Modularity	0.864	0.846
Max k-core	16	15

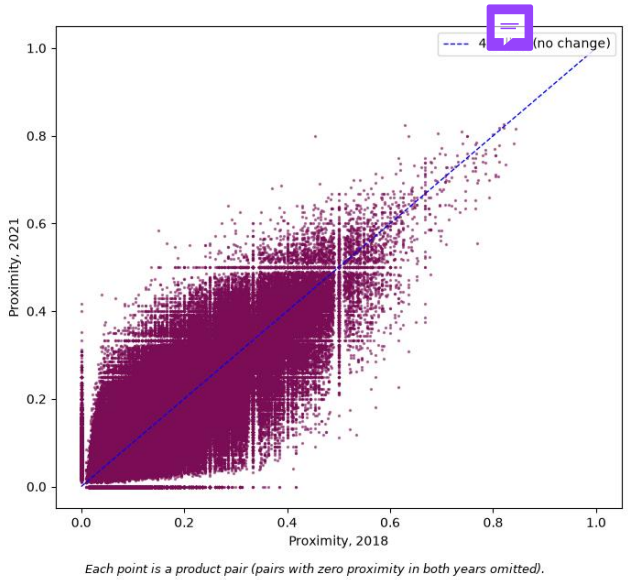
1. Table: Descriptive statistics of the product space networks, 2018 and 2021

	All product pairs	Pairs with non-zero proximity in at least one year
Pearson	0.8161	0.8023
Spearman	0.8025	0.787

2. Table: Correlation of edge weights (proximity) between 2018 and 2021

The correlation table shows that the similarity between product pairs is very consistent throughout the two years, meaning that the structure of the product space is stable. If we filter out the product pairs that are zero in both years, the correlation decreases slightly but still shows a strong, positive relationship. The correlation coefficients for Pearson and Spearman are close to each other, suggesting that both the exact values (Pearson) and the order of the relationships (Spearman) are similarly stable.

Although the edge weights are highly correlated across the two years, only 46.7% of the pairs classified as strong in 2018 are still above the 0.55 threshold in 2021. This is partly a threshold effect: many strong links lie close to the cutoff, so small changes in proximity are enough to move them below it.



3. Figure: Proximity of product pairs in 2018 and 2021

In Figure 3, we can see that the data points cluster around the diagonal; thus, relationships that were strong in 2018 tend to remain strong in 2021 as well. This is consistent with a correlation coefficient of around 0.8. However, there is significant variation around the diagonal; at higher values (above 0.5), the data points are sparser and more scattered, so stability is lower for individual pairs in the case of strong relationships. This is consistent with the fact that 46.7% of strong relationships remain strong. Bands along the axes indicate pairs whose relationships appeared or disappeared each year. Their values are mostly low (below 0.4), so it is primarily weak relationships that break down or form.

AI declaration: For this assignment, I used DeepL for translation, Grammarly to improve grammar and sentence structure, and Gemini to generate R plots (prompt history attached).